XrayAtlas: Automated X-Ray Classification and Image Registration

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Abstract. This report presents an end-to-end system for X-ray image registration and classification that can help improve efficiency and accuracy in X-ray image workflows in medical settings. Our project aimed to accurately classify X-ray images into their anatomical regions and was made up of three primary parts: Conversion of DICOM images to PNG format, CNN based classification system with multi-label classification, and full body registration that maps the classified images onto a full body anatomical template for visualization. Our system's classification accuracy was improved with batch normalization, hyperparameter optimization, and other improvements detailed in the report. The registration system uses image registration in order to accurately map the classified images onto the full body template. Although there were several challenges with classification accuracy and template creation, our system demonstrates the potential of combining deep learning with image registration for the classification of X-ray images.

Keywords: Biomedical Image Processing · Image Classification · Image Registration · X-ray Imaging · Template Alignment · Clustering · Python · OpenCV

1 Introduction

Medical imaging plays and important role in modern healthcare, with X-ray imaging remaining one of the most widely used tools due to its accessibility, cost effectiveness, and its value as a diagnostic tool. The accurate classification of X-ray images is an important step within medical image analysis pipelines as it allows for the correct image routing within the larger clinical system.

1.1 Problem Description and Motivation

Although there have been significant advances in the field of medical imaging, the automation of the classification of X-ray images based on their anatomical region remains challenging for several reasons. X-ray images can be variable in both position and contrast, which is amplified when comparing images across anatomical regions and from a variety of imaging tools. Additionally, many X-ray

images show more than one anatomical region at the same time requiring multi-label classification approaches rather than a more simple single label method. In a clinical setting, the accurate classification of X-ray images is very important as the misclassifications of X-ray images can lead to errors with detrimental consequences, potentially leading to delayed or incorrect diagnosis, and increases radiologists workload. Therefore, an accurate and efficient classification system would help to streamline the workflow, which would reduce the potential for human error and potentially act as a stepping point for future development in computer diagnostic tools.

1.2 Background and Related Work

Convolutional Neural Networks (CNNs) have emerged as the dominant approach for medical image classification. Early works by Rajpurkar et al. demonstrated that deep learning models like CheXNet could achieve near radiologist level performance in detecting pathologies from chest X-rays [**?**]. Modern CNN architecture combined with transfer learning have become standard practice for medical image classification tasks, using large datasets like ChestX-ray14 and CheXpert [**?**].

A key challenge in X-ray interpretation is multi-label classification, as a single X-ray image can show more than one anatomical region. In order to address this, recent approaches have begun to use anatomical region information into classification models. Agu et al. developed a system that detects anatomical sub regions using a Faster R-CNN before applying a graph convolutional network which relates findings across regions [**?**]. Methods like this help to improve the accuracy of diagnosis [**?**].

Researchers have also been exploring anatomical registration methods to align X-ray images to standard templates, enabling more consistent analysis across different images [**?**]. Such advances in image alignment, combined with deep learning classification techniques, are promising as they have the potential to enhance integration with clinical workflow and improve the accuracy and reliability of diagnoses.

The rest of the report is organized as follows. Section 2 describes the materials used in our study, including the X-ray dataset and the anatomical classifications. Section 3 presents our methodology, including the classification pipeline and the full body registration. Section 4 shows the qualitative and quantitative results of our implementation, both classification metrics and visualization examples. Section 5 discusses our accomplishments and the challenges encountered during the development. Section 6 outlines the specific contributions made by each team member. Section 7, presents a conclusion to our findings, and section 8 suggests potential directions for future work building upon our system.

2 Materials

2.1 X-Ray Dataset

Our project utilized a medical X-ray dataset which was comprised of X-ray images of various anatomical regions of the body. The dataset was orignally provided in DICOM format with CSV files containing image identifiers and anatomical labels.

2.2 Anatomical Classes

The dataset contains images of various anatomical regions, categorized into 22 anatomical classes. Table 1 presents 22 anatomical classes with their numeric classifications.

Table 1. Primary Anatomical Classes and Their Integer Classifications

Class ID	Anatomical Region	Class ID	Anatomical Region
0	Abdomen	11	Knee
1	Ankle	12	Lower Leg
2	Cervical Spine	13	Lumbar Spine
3	Chest	14	Others*
4	Clavicles	15	Pelvis
5	Elbow	16	Shoulder
6	Feet	17	Sinus
7	Finger	18	Skull
8	Forearm	19	Thigh
9	Hand	20	Thoracic Spine
10	Hip	21	Wrist

3 Methods

3.1 X-ray Classification Pipeline

We developed a pipeline for X-ray image classification which consisted of several stages: data preprocessing, model architecture design, training and evaluation. This section explains the process used to classify X-ray images into anatomical categories.

Data Preprocessing The classification system begins with the preprocessed PNG images. In this stage each X-ray image undergoes several preprocessing steps:

1. **Resizing:** All images are standardized to be 128×128 pixels, which ensures consistency across all images in the system.

2. **Normalization:** The pixel values are normalized with contrast normalization(CV2.NORM_MINMAX) t0 $[0, 500]$which helps improve the visibility of features and ensures that there is consistent contrast.
3. **Channel Dimension:** A channel dimension is added to each image giving each image the shape $(128, 128, 1)$.

Label Processing The dataset has 22 anatomical classes which represent the X-ray images, however, several images belong to more than one class. In order to address this several strategies were employed:

1. **Label Encoding:** The anatomical classes were each given a unique numerical label.
2. **1-to-K Encoding:** In order to address images with more than one class, the categorical labels were transformed into binary matrix representation using pandas' get_dummies function, creating 41 classes each represented by a vector.

Data Augmentation In order to improve the model's generalization we implemented data augmentation to account for variation within the images using TensorFlow's image augmentation tools:

1. **Random Rotation:** Images are rotated by up to $\pm$10degrees.
2. **Random Zoom:** $\pm$10% zoom is applied to the image
3. **Random Translation:** A vertical and/or horizontal shift of up to 10%.

The data augmentation is applied during the training process with 50% probability for each image within the training process. This increased the diversity of images in the training set without loosing the representation from the original images.

Model Architecture We implemented a convolutional neural network using the following architecture:

Key elements in the architecture include:

1. **Batch Normalization:** Added after each convolutional layer to speed up convergence and ensure stable training.
2. **Multiple Dropout Layers:** Multiple dropout layers with varying rates between 0.3- 0.5 ensured effective regularization.
3. **Sigmoid Activation:** used in the output layer for multi-label classification.

Training The training strategy employed used multiple stages which ensured the best possible results.

1. **Loss Function:** Binary Cross Entropy was used as the loss function
2. **Optimizer:** An Adam Optimizer with a specified learning rate of 0.001.

3. **Class Weighting:** Class weights were assigned based on the number of images in each class, helping to make the representation training process more equal across all anatomical classes.
4. **Early Stopping:** Use of patience of 30 epochs helped to limit overfitting.
5. **Model Checkpoint:** Ensured that the best model was saved based on the validation loss.

Prediction and Validation For model prediction and evaluation we followed the following steps:

1. **Threshold Based Classification:** Applied a threshold of 0.5 to the sigmoid output to determine class.
2. **Default Classification:** When no class probabilities above the threshold were identified, the system defaults to the highest prediction score.

The classification system is an important component of out X-ray analysis pipeline, as it provides classified X-ray images to the registration system.

3.2 Full Body X-ray Template Creation

Initial Approach Initially, our idea was to construct a comprehensive full-body X-ray template directly from a provided dataset of individual X-ray images. The steps involved were:

1. **Sorting the Dataset:** First, the images from a dataset (already converted to .png from .dcm) were sorted into directories based on their anatomical labels (e.g., "Chest", "Hand", "Knee").
2. **Clustering Images:** As there are many images capturing different body parts from different sides, each anatomical site's X-ray images were grouped to identify and select the most representative images using the KMeans clustering algorithm. This step aimed to reduce noise and variability, ensuring only high-quality, consistent images would form the basis for the template.
3. **Image Registration and Selection:** Selected images from each cluster underwent a two-stage robust registration process. Initially, each image was registered to a fixed reference image using mutual information as a similarity metric. Based on registration quality metrics, only the top images were retained for further processing. A second registration pass refined alignment accuracy by registering these top images to a preliminary averaged template.
4. **Template Averaging and Stitching:** The registered images from each anatomical region were then averaged using median pixel intensity to create per-site templates. Finally, these individual templates were stitched into a cohesive full-body template, based on predefined anatomical positions.

Revised Approach (Final) Upon reviewing out initial method, the professor indicated this approach was incorrect. Averaging diverse anatomical X-rays would not yield anatomically accurate or diagnostically reliable results due to

variability across images. Instead, the professor recommended starting from an existing full-body X-ray template found online. For this approach we followed this steps:

1. **Finding and Dividing a Suitable Template:**For our goals, we found a high-quality full-body x-ray image in Google. As it is just a regular image, we manually divided it into different parts for each anatomical region according to the classification labels from the dataset, such as "Chest", "Hip", "Knee", etc. Then we carefully mapped out each region with precise bounding boxes.
2. **Registration:** In order to properly register the x-ray image onto the full-body template, we first used the predicted labels from the model to find the area of the template that would be searched. We used mutual information as an objective function that was maximized using a multi-resolution strategy with gradient descent optimization, applying a similarity transformation on the x-ray image. For our search parameters, if the body part label is an extremity, we allow for reflection along the x- and y-axis. Otherwise, we only allow for a reflection along the y-axis to account for a mirrored image.
3. **Overlay:** After the X-ray image is transformed, we blend it onto the full-body template as a final output.

4 Results

4.1 Classification Performance

Figure 1 shows the training progress of our classification model of 200 epochs. The model achieved a final validation accuracy of 77.8% and a training accuracy of 99.2%. As evidenced by the graph a significant performance jump occurred around epoch 100, where the validation accuracy improved from approximately 55% to over 70%. The loss curve indicates that our approach effectively minimized both training and validation loss, with validation loss stabilizing after 100 epochs. The gap between the training and validation accuracy suggests that there was some degree of overfitting, although validation metrics remained stable through later epochs.

4.2 Visualization Results of Registration

The registration of the X-ray images onto a full-body template came with mixed results. For body parts with distinct features, the registration does a good job. However, if the body part of an X-ray image has a different orientation than what is shown the limits of similarity transformations will result in unsatisfactory results. This could also be attributed to the template, which has proportions different from the X-ray images we used. The X-ray images also suffer from clarity and noise from soft tissues and organs.

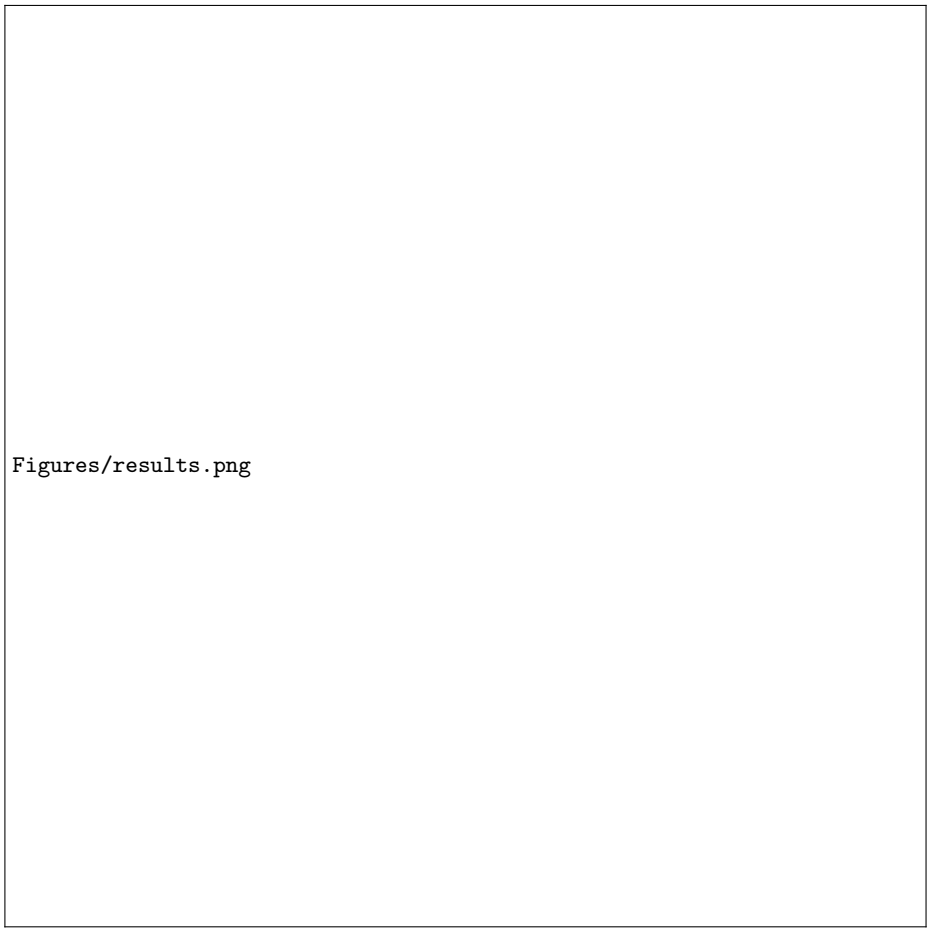


Fig. 1. Classification accuracy and loss for validation and training

5 Accomplishments

5.1 Mykhailo Isyp

Throughout the course of this project, I developed a deeper understanding of several key concepts in image analysis. Working on this project provided hands-on experience with techniques such as KMeans clustering, PCA for orientation correction, and robust image registration.

I have also learned the role of selecting an appropriate template. Our initial attempt to construct a template by averaging dataset images showed me the complexity of choosing the appropriate images automatically and trying to analyze and put them in the right place. During this process, I have overcome many obstacles. My efforts to create a unified template from scratch were met with

challenges due to inconsistencies in image positioning, orientation, and coverage across a dataset. Resolving these required multiple iterations of image clustering, registration, and alignment strategies.

After receiving feedback from the professor, the focus was shifted from building a template from scratch to using an existing one found online. It has significantly improved our output quality and I have learned how to define the bounding boxes for each body part and overlay images on it.

5.2 Niall Shaidullin

Throughout the project, I enhanced my knowledge of image processing, visualization, and plotting in medical imaging. I learned how to convert images between different formats, organize X-ray images based on anatomical labels, stitch the images into the template, and the basics of image registration.

I accomplished developing a full-body X-ray visualization, applying knowledge on image registration from the course. It required fixing alignment and overlaps, template sewing, and the site location logic. Working with real medical data gave me experience in standardizing input images.

In general, this project helped me improve my code ethic and collaboration skills. Using git as a version control tool, splitting the code into modules, and writing the report gave me valuable experience in team development.

5.3 Alina Pharis

Throughout this project, I deepened my understanding of deep learning approaches for medical image classification. Working on the classification component provided hands-on experience with neural network architecture design, hyperparameter tuning, and handling imbalanced datasets in a medical context. I focused on improving the classification system by adapting the original codebase with several enhancements. This involved implementing batch normalization layers for training stability, and optimizing dropout rates to prevent overfitting. One of the most challenging aspects was addressing the multi-label nature of X-ray images, where a single radiograph often contains multiple anatomical regions. I overcame this by developing a 1-of-K encoding system and implementing class weighting to handle the imbalanced distribution of anatomical classes. These improvements collectively raised the model's accuracy by over 5%. Despite these successes, I encountered persistent challenges with the train-test performance gap, suggesting dataset biases or labeling inconsistencies in the original data. Learning to diagnose and address these real-world data issues gave me valuable insight into the practical challenges of applying deep learning to medical imaging beyond theoretical models.

5.4 Eric Leeson

This project helped enhance my understanding of deep learning as well as its various applications within the realm of biomedical image classification. By working

on the classification code, I was able to get hands-on experience with tweaking the different model parameters, as well as different machine learning libraries such as TensorFlow. Additionally, by working on the registration code, I got a better idea of the challenges that come with accurate image alignment. There were many trade-offs involved with registration, as the more accurate the alignment was, the more time it took for the code to register it. Additionally, there were certain limitations when using a generic 2D template for the skeleton. For example, it was difficult to register differently oriented body parts such as hands due to the lack of dimensions on the template.

5.5 Ziqian Fan

Over the course of this project. I have learned to better use machine learning algorithms and tools for biomedical purposes. Referencing a classification model online and implementing it for our project, I played with the data and code to better understand how the model behaves and functions. This also helps to learn TensorFlow, as I had only experience with PyTorch. I also assisted with registration, using the SimpleITK framework to help implement and optimize the similarity transformation. Using online resources, I have overcame the challenge of integrating the processes together, allowing them to work with each other. However, the challenge I did not overcome was the hurdles that came with the registration portion, as certain body parts are difficult to register to the template due to clarity and orientation issues.

6 Contributions

6.1 Mykhailo Isyp

I developed the initial Python code for key functions such as `register_new_xray`, `load_xray_images`, `register_image`, `create_xray_template`, `save_template`, `display_image`, and `compute_difference_map`, which formed the basis for generating a basic X-ray template. This version was later refined by Niall. Once I received his improved version, I found that it still did not produce the desired full-body result. As a result, I completely rewrote the logic for full-body template construction (as implemented in `full_body_template.py` in the March 27 commit) and created `run.py` to execute this pipeline.

Afterward, I shared the updated version with Niall, who worked on improving the overlaps and anatomical site positioning. Once I received the updated implementation, I further refined the output by removing overlapping regions, rotating misaligned hands, cropping duplicate images, and cleaning up the codebase, organizing it into separate files and removing redundant functions.

Following feedback from the professor, we pivoted from constructing the template from dataset images to using a pre-existing full-body X-ray template found online. I found a suitable template, manually divided it into anatomical regions based on our class labels, and implemented the registration and overlay logic.

This version was then handed off to Eric, who worked on further optimizing the registration process.

I also wrote the script for the Demo Video, did a voice over using ElevenLabs and wrote README.md.

6.2 Niall Shaidullin

I implemented features for template construction and visualization of the anatomical site data. I implemented functionality for converting DICOM files to PNG. I worked closely with Mykhailo, who implemented the basis for template creation. I enhanced the full-body stitching logic, enhanced the run.py, and implemented functionality for visualizing label positioning for testing purposes. As the project developed, I enhanced the label positioning and implemented functionality to allow combined labelling so the template highlights multiple regions based on the input x-ray image.

I worked on the functionality for highlighting the needed part of the x-ray template for the desired output. This involved outlining the specific label based on the model's predicted output. This made it easier to interpret the result.

Throughout the project, I attended every team meeting, shared updates and thoughts on the project development, and contributed to task coordination. I contributed to the report writing and finalizing the pipeline for demonstration.

6.3 Alina Pharis

I began with the original classification code which was developed by Ziqian and implemented several significant improvements which enhanced the performance and functionality. My primary contributions include adapting the code to address key performance limitations through the addition of batch normalization layers after each convolutional layer which ensured that the training remained more stable and improved the convergence. In addition to this I restructured the original model architecture to include more effective dropout implementation with rates between 0.3-0.5 which helped to prevent over-fitting. Additionally, I also added more comprehensive data augmentation including controlled rotations ($\pm$10 degrees), zooms($\pm$10%) and translations ($\pm$10% both horizontally and vertically) which improved model generalization.

A major contribution was the addition of a multiphase training strategy, which allowed for precise parameter update in early phases while enhancing computational efficiency in later stages. I also implemented class weighting which helped to ensure that the training was equal across all different anatomical regions. I then pushed the classification code to the github repository.

For the written report, I wrote the sections which explained the classification methodology, which included certain elements of data preprocessing, model architecture modifications and training strategies. I also contributed to the materials section by formatting and organizing information about the dataset characteristics and anatomical classes, and contributed to the accomplishments sections relating to the classifications section.

6.4 Eric Leeson

In this project, I helped work on both the classification and registration code. For the classification, I helped decide on a dataset and similar project to use as reference, as well as testing and training the classification model. My main contribution was the registration code. I received the original version of the iteration code from Mykhailo and Niall, which did not include the use of an objective function that would be optimized.

From there, I followed instructions from the professor on how to implement the brute-force search method to maximize the normalized cross-correlation between the image and the template using the OpenCV library. Using the code provided by Mykhailo and Niall, I was able to also blend and display the final image.

6.5 Ziqian Fan

I developed the initial Python code for classification, referencing the code submitted for the Kaggle competition on the X-ray dataset that we used. I also wrote the initial code for converting Dicom to PNG. With the assistance of the ChatGPT and SimpleITK documentation, I was able to use the framework to improve the current registration code, using an optimization algorithm instead of the brute-force method that Eric Leeson implemented. I also adapted some of the helper functions written by Nial and Mykhailo to fit the image type that I was working with.

7 Conclusion and Discussions

In this project, we developed an end-to-end system for the classification and visualization of X-ray images through anatomical registration on a skeletal template. This approach was divided into 2 key parts, the classification of the x-ray images by their anatomical region, and the registration of x-rays onto a standardized template.

The classification component had promising results with the final validation consistently showing an accuracy between $77 - 80\%$, due to the integration of batch normalization layers, strategic dropout implementation, and multiphase training. The training progression showed a significant increase in performance around epoch 100 which suggested to us that deeper networks, if given enough time can effectively learn to identify smaller differences between anatomical regions.

The registration component was effective for certain anatomical regions for example the chest, with near perfect alignment, however, there remained some errors when positioning extremities, for example the hands or feet, where the x-ray images showed much more significant variation which made positioning them more challenging. A key limitation here was that our template only represented the anatomy from one orientation, while the x-ray images we were attempting

to project onto it were from numerous different angles, which limited our ability to properly visualize it. Additionally, our region mapping was fixed to a specific side of the body which made certain body parts like shoulders and feet variable in the quality of their representations. Both of these issues would be areas which could be improved in the continuation of this project.

Our integration approach demonstrates that it is possible to create a system to classify X-ray images and visualize them. This has potential applications across the medical and clinical fields, in radiology, and potentially even in computer diagnosis tools. By presenting the X-ray findings through X-ray registration, it allows for a more intuitive way to interpret X-ray findings, beyond just classification. Despite these achievements, there are still several limitations. The gap between training and validation accuracy indicates potential over-fitting, which could be addressed through increasing the training data size or increased regularization. The registration accuracy varies significantly across different anatomical regions, which indicates to us that it may be necessary to customize the solution for each body part. Despite these limitations however, the deep learning approach for X-ray categorization in combination with the registration component form a solid foundation which can be build upon in order to create a more sophisticated medical image analysis system.

8 Future Work

As a team, we developed an end-to-end software for X-ray site classification with the template visualization, there are still several promising avenues for future enhancement. We have identified the following directions which have the potential to expand the functionality and improve the system for future works

8.1 Dataset Generalization and Expansion

To increase the system's applicability in more environments, we suggest generalizing the pipeline beyond the Unifesp dataset, and training on additional X-ray datasets. Which can increase the robustness of our application by training classification and visualization and help to identify and mitigate any biases unique to the dataset.

8.2 Template

In the end, we used a full-body X-ray image found online, but we believe there is an opportunity to stitch the images from the dataset into a clear and sophisticated picture to use as a template. By using this as an alternative the customized template would more accurately represent variation in human anatomy and potentially increase the accuracy of the registration.

8.3 Confidence Scoring

Finally, our current version of the pipeline does not illustrate the model's prediction accuracy of anatomical regions. By integrating the a confidence score in the visualization, it would help radiologists and other clinical staff to better interpret the results and help to identify where weak classifications exist.

Through the exploration of these directions in future works and studies, our X-ray classification and visualization system has the potential to become a more comprehensive tool and expand its potential use in the medical and clinical space.

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References